Practical 10 Iron-rich rocks

In this practical we'll look at the petrography and geochemistry of different Fe-rich rocks. This includes banded iron formation (BIF) manganese formation (MnF), and oolitic ironstone. They are all sedimentary iron-rich rocks, where primary iron accumulated in a marine setting.

1) Petrography

1.1 BIF

There are two hand samples for BIF. In addition, a typical photograph of a core is below (Fig 1). Look at the hand samples/photographs and describe your observations:

- color, color variation
- bedding and scale of bedding
- grain size
- any other features of note

1.2 MnF

In the absence of actual hand samples of MnF in the department, use this link to access hand sample and thin section images of a manganese formation:

Manganese ore

Describe your observations:

- color, color variation
- bedding and scale of bedding
- grain size
- any other features of note

1.3 Oolitic ironstone

Again, in the absence of actual hand samples of oolitic ironstones in the department, use this link to access hand sample and thin section images of an example:

Oolitic ironstone

Describe your observations:

- color, color variation
- bedding and scale of bedding
- grain size, grain types
- any other features of note

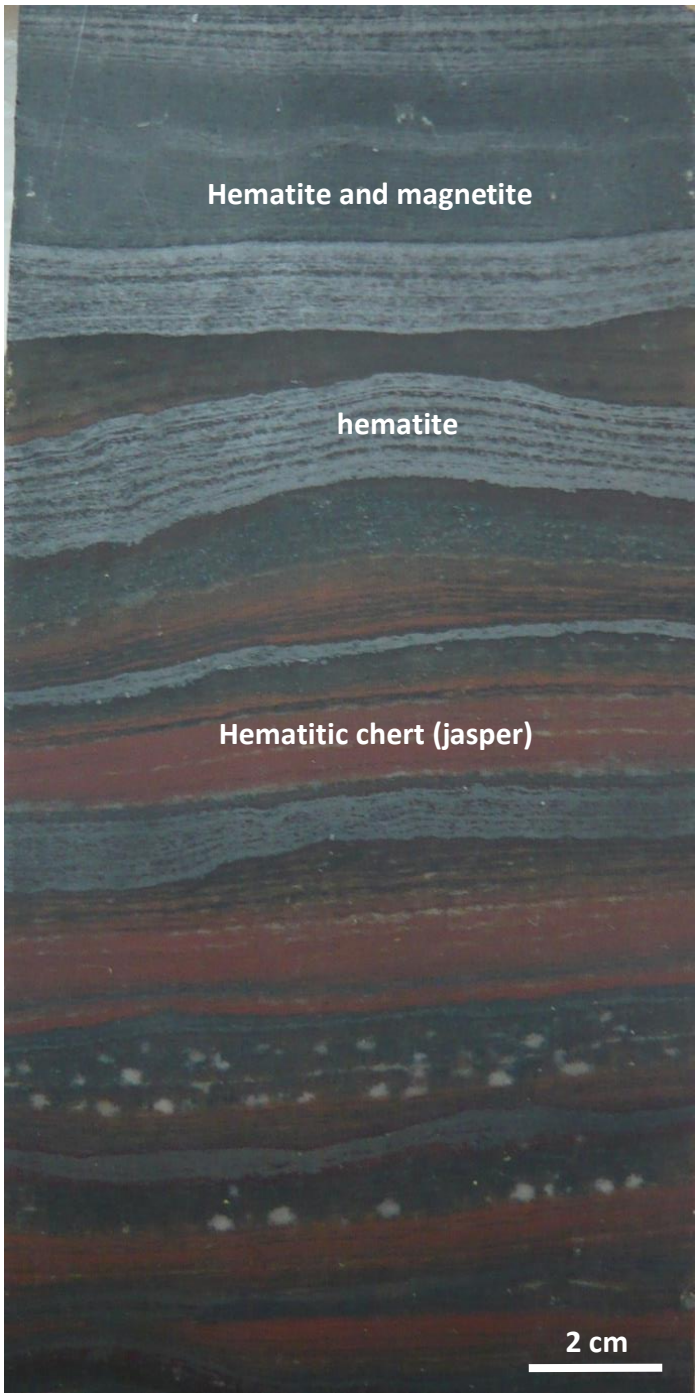


Fig. 1: Core photograph of banded iron formation
cm-scale interbedding of various lithologies, each internally laminated at mm scale; grain size: very fine (Fe-rich mud); white dots at base are secondary carbonate nodules

1.2 MnF

Observations are similar to BIF: bedding/lamination, fine grain size
Note that both BIF and MnF have been (in this case) overprinted by later metamorphism, hence the garnet in 1.2

1.3 Oolitic ironstone

Fe ooids, some with quartz nucleus, quartz grains, fossil shells, medium to fine sand grain size, calcite cement; higher energy for deposition

2) Geochemistry

For the BIF and oolitic ironstone estimate the percentage of iron-bearing minerals in the sample (iron minerals in BIF, ooids in oolitic ironstone). Calculate the amount of pure iron that would be contained in $1m^3$ of each sample.
Assume the following typical mineralogy:
BIF: hematite Fe_2O_3
Oolitic ironstone: siderite $FeCO_3$

Use the percentage estimation chart at the back (Figure 2).

Molar weights:
Fe: 55.85 g/mol
O: 16 g/mol
C: 12.01 g/mol

Density of hematite: 5.26 g/cm^3
Density of siderite: 3.96 g/cm^3

Answer:
Hematite: (2x55.85) + (3x16) = 159.7; 2x55.85=111.7/159.7 = 69.9% Fe
Siderite: 55.85 + 12.01 + (3x16) = 115.86; 55.85/115.86 = 48.2% Fe

Assume a $1m^3$ sample with 30% hematite (determined from thin section)
0.3 m^3 hematite = 0.3 x 10^6 cm^3
Density hematite: 5.26 g/cm^3
i.e. sample contains 5.26 x 0.3 x 10^6 = 1,578,000 g = 1,578 kg hematite
of it 69.9% Fe: 1,578 kg x 0.699 = 1,103 kg Fe

Assume a $1m^3$ sample with 30% (-40%) siderite (determined from thin section)
0.3 m^3 siderite = 0.3 x 10^6 cm^3
Density siderite: 3.96 g/cm^3
i.e. sample contains 3.96 x 0.3 x 10^6 = 1,188,000 g = 1,188 kg siderite
of it 48.2% Fe: 1,188 kg x 0.482 = 572.6 kg Fe

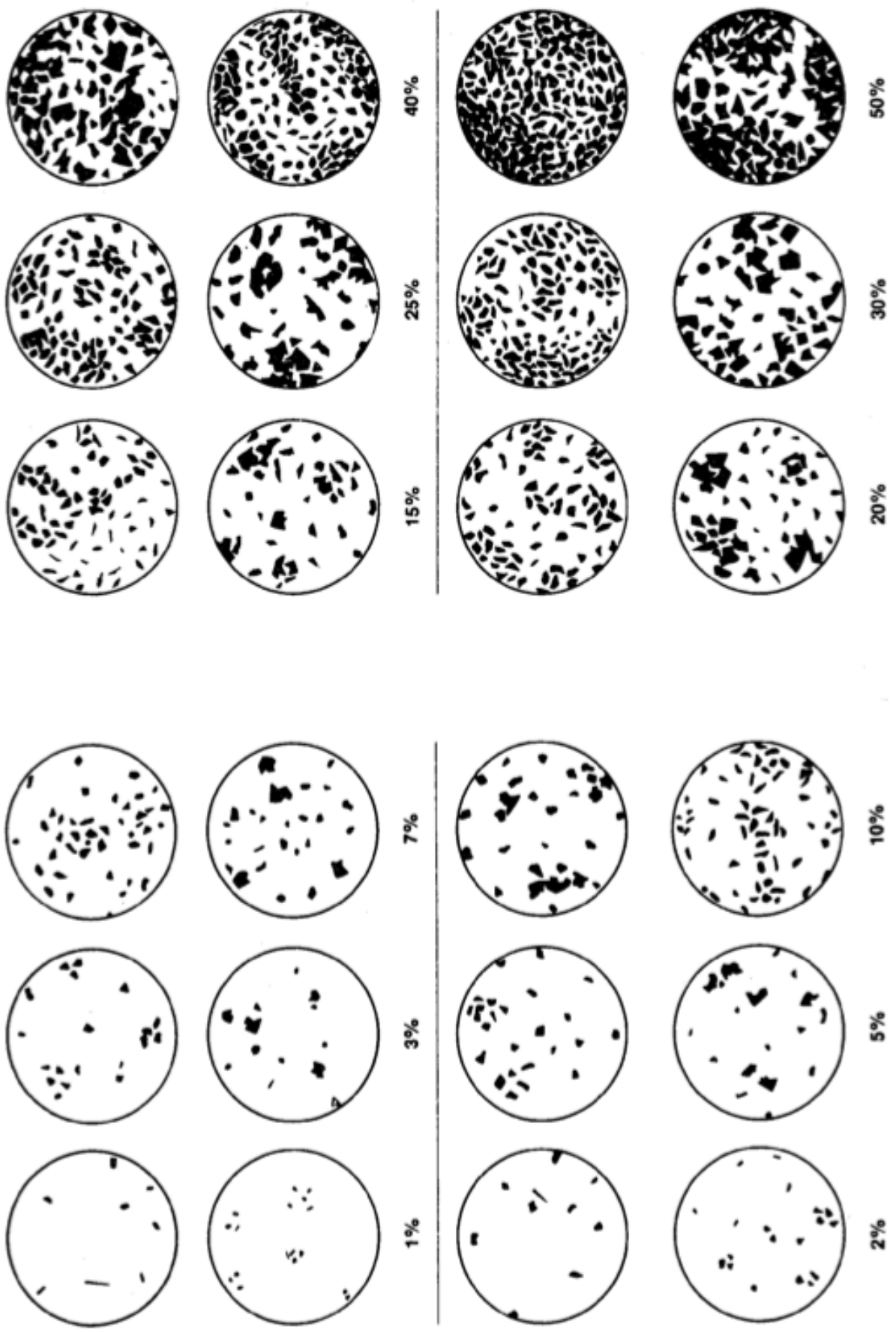


Figure 2: Chart to estimate percentage of components in a rock/thin section.